

Part I

SUMMARY of the Fundamental Theorems of Calculus

Summary

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The First Fundamental Theorem of Calculus:

Given a continuous function f on the interval $[a, b]$, we define an accumulation function, A by

$$A(x) = \int_a^x f(t) dt$$

The First Fundamental Theorem of Calculus (FFTOC) states that

$$A'(x) = f(x)$$

or, equivalently, that

$$\frac{d}{dx} \int_a^x f(t) dt = f(x)$$

Note 1: $A(a) = 0$

Note 2: The function A is an antiderivative of f .

Therefore, any antiderivative, G of f can be written as $G(x) = A(x) + C$, or, equivalently,

$$G(x) = \int_a^x f(t) dt + C.$$

Since $G(a) = A(a) + C = 0 + C = C$, it follows that $G(x) = A(x) + G(a)$, or, equivalently, that

$$G(x) = G(a) + \int_a^x f(t) dt$$

Note 3: Let f' , the derivative of f , be continuous on $[a, b]$. Since f is its antiderivative, we have

$$f(x) = f(a) + \int_a^x f'(t) dt$$

The Second Fundamental Theorem of Calculus:

FFTOC implies that, for $x = b$, $A(b) = \int_a^b f(t) dt$. Since $A(a) = 0$, we can write

$$\int_a^b f(t) dt = A(b) - A(a)$$

Note 4: This is the Second Fundamental Theorem of Calculus (SFTOC).

Note 5: Equivalently, using the result in Note 3, for $x = b$, $f(b) = f(a) + \int_a^b f'(t) dt$. So $\int_a^b f'(t) dt = f(b) - f(a)$, or equivalently,

$$\int_a^b \frac{d}{dx} f(x) dx = f(b) - f(a).$$

Part II

Recitation Questions

Problem 1

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Problem 1

Problem 1 True or False: If f is continuous on the closed interval $[a, b]$, then

$$\frac{d}{dx} \left(\int_a^b f(t) dt \right) = f(x)$$

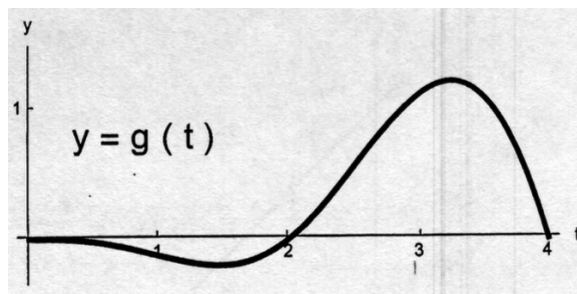
Explanation. False. $\frac{d}{dx} \left(\int_a^b f(t) dt \right) = 0$ since $\int_a^b f(t) dt$ is a constant and $\frac{d}{dx}(\text{constant}) = 0$.

Problem 2

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Problem 2

Problem 2 The graph of g , a continuous function on $[0, 4]$, is shown in the figure. Let $A(x) = \int_0^x g(t) dt$, for $0 \leq x \leq 4$.



(a) Circle the correct statement about $A(2)$.

- (i) $A(2) = 0$;
- (ii) $A(2) > 0$;
- (iii) $A(2) < 0$;
- (iv) none of the previous answers.

Explanation. Correct choice: (iii) $A(2) < 0$. The curve $g(t)$ on the interval $(0, 2)$ is below the x -axis and therefore the net area is negative.

(b) Circle the correct statement about $A(3.8)$.

- (i) $A(3.8) = 0$;
- (ii) $A(3.8) > 0$;
- (iii) $A(3.8) < 0$;
- (iv) none of the previous answers.

Explanation. Correct choice: (ii) $A(3.8) > 0$. The net area under the curve $g(t)$ on the interval $(2, 3.8)$ is positive and larger than the negative net area on the interval $(0, 2)$. Therefore, the net area on the interval $(0, 3.8)$ is positive.

(c) Circle the correct statement about $A'(3.8)$.

- (i) $A'(3.8) = 0$;
- (ii) $A'(3.8) > 0$;
- (iii) $A'(3.8) < 0$;
- (iv) none of the previous answers.

Explanation. Correct choice: (ii) $A'(3.8) > 0$ because $A'(3.8) = g(3.8)$

Problem 2

- (d) Find the solution of the following initial value problem: $y'(x) = g(x)$ and $y(0) = 2$.

- (i) $y(x) = g(x)$;
- (ii) $y(x) = g(x) + 2$;
- (iii) $y(x) = A(x)$;
- (iv) $y(x) = A(x) + 2$;
- (v) $y(x) = g'(x)$;
- (vi) $y(x) = g'(x) + 2$;
- (vii) none of the previous answers.

Explanation. Correct choice: (iv) $y(x) = A(x) + 2$. $y'(x) = A'(x) = g(x)$ and $y(0) = A(0) + 2 = 2$

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- (e) Find the expression for $\int_0^4 |g(t)| dt$.

- (i) $A(4)$;
- (ii) $A(2) - A(4)$;
- (iii) $A(4) - A(2)$;
- (iv) $A(4) - 2A(2)$;
- (v) none of the previous answers.

Explanation. Correct choice: (iv) $A(4) - 2A(2)$.

$$\int_0^4 |g(t)| dt = \int_2^4 g(t) dt - \int_0^2 g(t) dt$$

and

$$\int_2^4 g(t) dt = \int_0^4 g(t) dt - \int_0^2 g(t) dt$$

Therefore,

$$\int_0^4 |g(t)| dt = \int_0^4 g(t) dt - 2 \int_0^2 g(t) dt = A(4) - 2A(2)$$

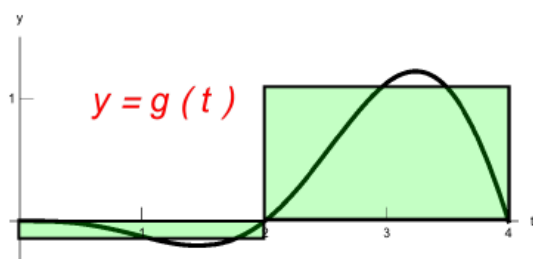
- (f) Find the midpoint Riemann sum for the function g on the interval $[0, 4]$ with $n = 2$ (the number of subintervals).

- (i) $g(1) + g(3)$;
- (ii) $g(0) + g(2)$;
- (iii) $g(2) + g(4)$;

Problem 2

- (iv) $2g(1) + 2g(3)$;
- (v) $2g(0) + 2g(2)$;
- (vi) $2g(2) + 2g(4)$;
- (vii) *none of the previous answers.*

Explanation. Correct choice: (iv) $2g(1) + 2g(3)$



Problem 3

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Problem 3 The limit of Riemann sums for a function f on the interval $[1, 5]$ is given by

$$\lim_{n \rightarrow \infty} \sum_{k=1}^n \left(2x_k^* + \frac{1}{x_k^*} \right) \Delta x \text{ on } [1, 5].$$

- (a) Identify f and express the limit as a definite integral.

Explanation.

$$f(x) = 2x + \frac{1}{x}$$

$$\int_1^5 \left(2x + \frac{1}{x} \right) dx = \lim_{n \rightarrow \infty} \sum_{k=1}^n \left(2x_k^* + \frac{1}{x_k^*} \right) \Delta x$$

- (b) Evaluate the limit of Riemann sums.

Explanation.

$$\int_1^5 \left(2x + \frac{1}{x} \right) dx = \left(x^2 + \ln |x| \right) \Big|_1^5 = (25 + \ln(5)) - (1 + \ln(1)) = 24 + \ln(5)$$

Problem 4

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Problem 4 Compute the following integrals:

(a) $\int_0^1 e^{5x} dx$

Explanation. $\int_0^1 e^{5x} dx = \left[\frac{1}{5} e^{5x} \right]_0^1 = \frac{1}{5} (e^5 - e^0) = \frac{1}{5} (e^5 - 1)$

(b) $\int_{-2}^{-1} \frac{1}{x^3} dx$

Explanation.

$$\begin{aligned} \int_{-2}^{-1} \frac{1}{x^3} dx &= \int_{-2}^{-1} x^{-3} dx \\ &= \left[\frac{x^{-2}}{-2} \right]_{-2}^{-1} \\ &= \left[\frac{-1}{2x^2} \right]_{-2}^{-1} \\ &= -\frac{1}{2} - \left(-\frac{1}{8} \right) \\ &= -\frac{1}{2} + \frac{1}{8} = -\frac{3}{8} \end{aligned}$$

(c) $\int_0^4 (3x - 5 + 7\sqrt{16 - x^2}) dx$

Explanation. First notice that:

$$\int_0^4 (3x - 5 + 7\sqrt{16 - x^2}) dx = \int_0^4 (3x - 5) dx + 7 \int_0^4 \sqrt{16 - x^2} dx. \quad (1)$$

To get the easy part out of the way first:

$$\int_0^4 (3x - 5) dx = \left[\frac{3}{2}x^2 - 5x \right]_0^4 = (24 - 20) - (0 - 0) = 4. \quad (2)$$

So the real issue is in computing $\int_0^4 \sqrt{16 - x^2} dx$.

We do not know how to integrate $\sqrt{16 - x^2}$ directly, but recall that the solution set to the equation $x^2 + y^2 = 16$ is a circle with radius 4 centered at the origin. Solving for y we get $y = \pm\sqrt{16 - x^2}$. So if we restrict

Problem 4

ourselves to $y = \sqrt{16 - x^2}$ and $0 \leq x \leq 4$, this gives us the upper-right quarter of the circle (draw a picture and convince yourself of this). Since the total area of the circle of $\pi(4)^2 = 16\pi$, the area under this portion of the curve is $\frac{1}{4}(16\pi) = 4\pi$. Thus we have computed (geometrically) that

$$\int_0^4 \sqrt{16 - x^2} \, dx = 4\pi. \quad (3)$$

So, $\int_0^4 (3x - 5 + 7\sqrt{16 - x^2}) \, dx = 4 + 7(4\pi) = 4 + 28\pi$ by (1),(2), and (3)

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Problem 5

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Problem 5 Find the derivative of the following functions:

$$(a) F(x) = \int_{\sqrt{x}}^1 \frac{t^2}{2 + 3t^4} dt$$

Explanation. First notice that

$$F(x) = \int_{\sqrt{x}}^1 \frac{t^2}{2 + 3t^4} dt = - \int_1^{\sqrt{x}} \frac{t^2}{2 + 3t^4} dt.$$

So we can apply the First Fundamental Theorem of Calculus, along with the chain rule, to compute:

$$F'(x) = \frac{d}{dx} \left(- \int_1^{\sqrt{x}} \frac{t^2}{2 + 3t^4} dt \right)$$

$$\begin{aligned} \text{we introduce the variable: } u = \sqrt{x} &\implies F'(x) = \frac{dF}{du} \frac{du}{dx} \\ &= - \frac{u^2}{2 + 3(u)^4} \cdot \frac{d}{dx}(\sqrt{x}) \\ &= - \frac{u^2}{2 + 3u^4} \cdot \frac{1}{2\sqrt{x}} \\ \text{substituting } u = \sqrt{x} &\implies F'(x) = - \frac{x}{2 + 3x^2} \cdot \frac{1}{2\sqrt{x}} \\ &= - \frac{\sqrt{x}}{2(2 + 3x^2)} \end{aligned}$$

$$(b) G(x) = \int_x^{x^3} \sin(7t^2) dt$$

Explanation. First notice that

$$\begin{aligned} G(x) &= \int_x^{x^3} \sin(7t^2) dt \\ &= \int_x^0 \sin(7t^2) dt + \int_0^{x^3} \sin(7t^2) dt \\ &= - \int_0^x \sin(7t^2) dt + \int_0^{x^3} \sin(7t^2) dt \end{aligned}$$

It is worth pointing out that breaking up the integral above at 0 was arbitrary. Since $\sin(7t^2)$ is continuous over $(-\infty, \infty)$, we could have chosen any real number in place of 0.

We can now apply the First Fundamental Theorem of Calculus, along with

the chain rule, to compute:

$$\begin{aligned} G'(x) &= \frac{d}{dx} \left(- \int_0^x \sin(7t^2) dt + \int_0^{x^3} \sin(7t^2) dt \right) \\ &= \frac{d}{dx} \left(- \int_0^x \sin(7t^2) dt \right) + \frac{d}{dx} \left(\int_0^{x^3} \sin(7t^2) dt \right) \end{aligned}$$

The left integral requires no chain rule on the FFTOC. $\frac{d}{dx} \left(- \int_0^x \sin(7t^2) dt \right) = -\sin(7x^2)$

The right integral requires chain rule. We'll substitute $u = x^3$.

$$\begin{aligned} \frac{d}{dx} \left(\int_0^{x^3} \sin(7t^2) dt \right) &= \frac{d}{du} \left(\int_0^u \sin(7t^2) dt \right) \frac{du}{dx} \\ &= \sin(7u^2) \cdot \frac{d}{dx}(x^3) \\ &= \sin(7u^2) \cdot 3x^2 \end{aligned}$$

$$\text{substituting } u = x^3 \implies \frac{d}{dx} \left(\int_0^{x^3} \sin(7t^2) dt \right) = 3x^2 \sin(7x^6)$$

Combining we have:

$$G'(x) = 3x^2 \sin(7x^6) - \sin(7x^2).$$

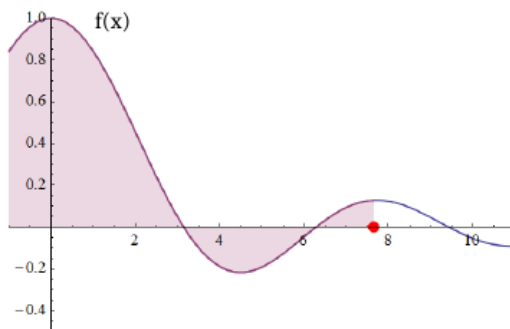
An alternative approach using SFTOC: Call $h(t) = \sin(7t^2)$ and let H be an antiderivative of h . By SFTOC, $\int_x^{x^3} \sin(7t^2) dt = \left[H(t) \right]_{t=x}^{x^3}$. Then:

$$\begin{aligned} \frac{d}{dx} \left(\int_0^{x^3} \sin(7t^2) dt \right) &= \frac{d}{dx} \left(\left[H(t) \right]_{t=x}^{x^3} \right) \\ &= \frac{d}{dx} (H(x^3) - H(x)) \\ &= H'(x^3) \cdot 3x^2 - H'(x) \\ &= h(x^3) \cdot 3x^2 - h(x) \\ &= \sin(7x^6) \cdot 3x^2 - \sin(7x^2) \end{aligned}$$

Problem 6

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Problem 6 Given the following graph of $y = f(x)$, let $g(x) = \int_{-1}^x f(t) dt$.



- (a) Is g continuous? Why or why not?

Explanation. $g(x)$ represents the area function (or the signed area between the curve $y = f(x)$ and the x -axis from -1 to x). By the FTC(I), $g(x)$ is continuous.

- (b) Is g differentiable? Why or why not?

Explanation. Similarly by the FTC(I) the function g is differentiable, and moreover:

$$\frac{d}{dx}(g(x)) = \frac{d}{dx} \left(\int_{-1}^x f(t) dt \right) = f(x)$$

- (c) Where does g achieve its absolute maximum and minimum values? Where does g achieve any local extreme values? Assume the domain of g is $[-1, 7.6]$.

Explanation. Since g is continuous on the closed interval $[-1, 7.6]$, g attains its absolute extreme values at either critical points or endpoints. But we just saw that $g'(x) = f(x)$, and so the critical points of g are just the points where $f(x) = 0$. Thus, the critical points of g are approximately 3.1 and 6.25.

Since $f = g'$, g is increasing when f is positive and g is decreasing when f is negative. So we have that $x = 3.1$ is a local maximum while $x = 6.25$ is a local minimum of g . This takes care of the local extreme values.

To find the absolute extreme values of g , we need to determine what are the largest and smallest values among $g(-1)$, $g(3.1)$, $g(6.25)$, and $g(7.6)$. And do not forget the $g(x)$ denotes the (signed) area of the region bounded by the curve and the x -axis.

Problem 6

Clearly $g(-1) = 0$ since a line segment does not have any area. At any time after $x = -1$ we have a positive (net) area. Thus, our absolute minimum occurs at $x = -1$. To find the absolute maximum of g , we need to compare $g(3.1)$ and $g(7.6)$. We see that we have the greatest area under the curve at $x = 3.1$ because by the time we get to $x = 7.6$, the area we have subtracted off between $x = 3.1$ and $x = 6.25$ is greater than what we have added back on from $x = 6.25$ to $x = 7.6$. Therefore, our absolute maximum occurs at $x = 3.1$.

- (d) *Where is the graph of g concave up? Concave down?*

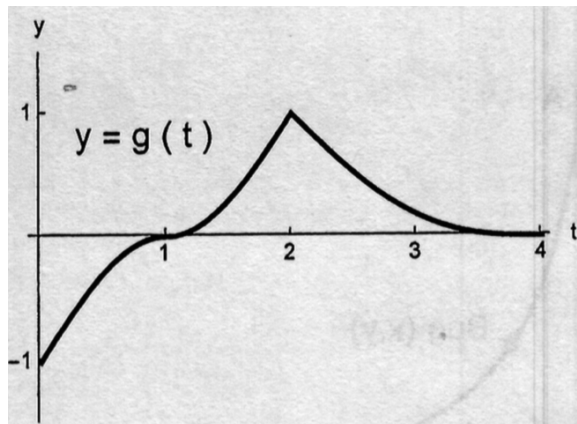
Explanation. g is concave up when f is increasing, which is on $(-1, 0) \cup (4.25, 7.6)$. Similarly, g is concave down when f is decreasing, which is on $(0, 4.25)$. Remember, $g' = f$.

Problem 7

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Problem 7 The graph of g , a continuous function $[0, 4]$, is shown in the figure.

Let $A(x) = \int_0^x g(t) dt$ for $0 \leq x \leq 4$.



(a) Circle the correct statement about $A(1)$.

- (i) $A(1) = 0$;
- (ii) $A(1) < 0$;
- (iii) $A(1) > 0$;
- (iv) none of the previous answers.

Explanation. Correct choice: (ii) $A(1) < 0$. $A(1)$ is the net area of the region bounded by the curve and the x -axis between $x = 0$ and $x = 1$. On this interval, $g(t)$ is negative and therefore, $A(1)$ is negative.

(b) Circle the correct statement about $A(1.5)$.

- (i) $A(1.5) = 0$;
- (ii) $A(1.5) < 0$;
- (iii) $A(1.5) > 0$;
- (iv) none of the previous answers.

Explanation. Correct choice: (ii) $A(1.5) < 0$. The net area on the interval $(1, 1.5)$ is positive and smaller than the negative net area on the interval $(0, 1)$. Therefore, the net area on the interval $(0, 1.5)$ is negative.

(c) Circle the correct statement about $A'(1.5)$.

- (i) $A'(1.5) = 0$;
- (ii) $A'(1.5) < 0$;

- (iii) $A'(1.5) > 0$;
- (iv) *none of the previous answers.*

Explanation. Correct choice: (iii) $A'(1.5) > 0$ because $A'(1.5) = g(1.5)$

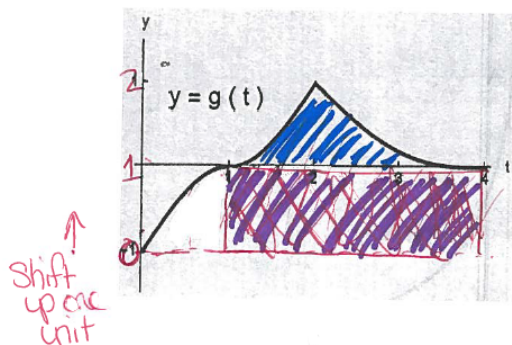
(d) Circle the correct expression for $\int_1^4 (g(t) + 1) dt$.

- (i) $A(4) + 1$;
- (ii) $A(4) - A(1)$;
- (iii) $A(4) - A(1) + 1$;
- (iv) $A(4) + 3$;
- (v) $A(4) - A(1) + 3$;
- (vi) *none of the previous answers.*

Explanation. Correct choice: (v) $A(4) - A(1) + 3$

We can split the integral: $\int_1^4 (g(t) + 1) dt = \int_1^4 g(t) dt + \int_1^4 1 dt = A(4) - A(1) + 3$

We can visualize this integral by imagining shifting the graph of g up one unit. See figure below. Our area we are looking for consists of the area under the curve g on the interval $(1, 4)$ (the blue region) plus the rectangular region formed below (the purple region).



The area of the blue region is $\int_1^4 g(t) dt = \int_0^4 g(t) dt - \int_0^1 g(t) dt = A(4) - A(1)$.

The area of the purple region is $\int_1^4 1 dt$. This integral is the area under the line $y = 1$ on the interval $(1, 4)$ which is a rectangular region of area 3.

Combining these, the total area $\int_1^4 (g(t) + 1) dt = A(4) - A(1) + 3$

(e) Circle the correct statement about $A(0)$.

- (i) $A(0) = 0$;
- (ii) $A(0) = 1$;
- (iii) $0 < A(0) < 1$;
- (iv) $A(0) > 1$;
- (v) $A(0) < 0$;
- (vi) none of the previous answers.

Explanation. Correct choice: (i) $A(0) = 0$. By the definition of A , $A(x) = \int_0^x g(t) dt$ for $0 \leq x \leq 4$. So, $A(0) = \int_0^0 g(t) dt = 0$.

(f) Circle the interval (or intervals) where the function A is DECREASING.

- (i) $(0, 1)$;
- (ii) $(1, 2)$;
- (iii) $(2, 4)$;
- (iv) none of the previous answers.

Explanation. Correct choice: (i) $(0, 1)$. A is decreasing means that the area is decreasing or we are accumulating negative area. This happens when A 's derivative, g is negative.

(g) Circle the value (or values) where the function A attains its MAXIMUM.

- (i) $x = 0$;
- (ii) $x = 1$;
- (iii) $x = 2$;
- (iv) $x = 3$;
- (v) $x = 4$;
- (vi) none of the previous answers.

Explanation. Correct choice: (v) $x = 4$. It appears that $A(x) < 0$ on $(0, 2)$ and $A(x) > 0$ on $(2, 4]$. So, the maximum occurs on $[2, 4]$. Since $A' = g$ is positive on $(2, 4)$, and A is increasing there, the maximum is attained at the end point, $x = 4$. In other words, A achieves a maximum when we have the largest positive area. Because we start with negative area and then continue to accumulate positive area until $x = 4$, the maximum occurs at the end point.

(h) Circle the value (or values) where the function A attains its MINIMUM.

- (i) $x = 0$;
- (ii) $x = 1$;
- (iii) $x = 2$;
- (iv) $x = 3$;
- (v) $x = 4$;
- (vi) none of the previous answers.

Explanation. Correct choice: (ii) $x = 1$. $A' = g$ is negative on $(0, 1)$ and positive on $(1, 4)$. So A is decreasing on $(0, 1)$ and increasing on $(1, 4)$. So the local and absolute maximum occurs at $x = 1$. In other words, A achieves a minimum when we have the largest negative area. We start by accumulating negative area and then switch to accumulating positive area at $x = 1$.

(i) Circle the interval (or intervals) where the function A is CONCAVE DOWN.

- (i) $(0, 1)$;
- (ii) $(1, 2)$;
- (iii) $(2, 4)$;
- (iv) none of the previous answers.

Explanation. Correct choice: (iii) $(2, 4)$. A is concave down where $A' = g$ is decreasing. g is decreasing on $(2, 4)$.

(j) Circle the value (or values) of x where the function A has an inflection point.

- (i) $x = 1$;
- (ii) $x = 2$;
- (iii) $x = 3$;
- (iv) none of the previous answers.

Explanation. Correct choice: (ii) $x = 2$. $A' = g$ changes from increasing to decreasing.

(k) Sketch the graph of A , based on items (e)-(j)

Explanation.

Problem 7

